Rock , paper, scissors game

Part one

Player 1:

1) specification:

- Simplify input 3 input , 2 output
- Rock =01 paper=10 scissors=11

2)truth table:

R1	p1	s1	A	B
0	0	0	0	0
0	0	1	1	1
0	1	0	1	0
0	1	1	0	0
1	0	0	0	1
1	0	1	0	0
1	1	0	0	0
1	1	1	0	0

3)A-map:

R1/P1 S1	00	01	11	10
0		1		1
1				

=R1`(P1 xorS1)

B-map:

R1/P1 S1	00	01	11	10
0		1		
1	1			

=P1(R1 xor S1)

Player 2:

1) specification:

- Simplify input 3 input , 2 output
- Rock =01 paper=10 scissors=11

2)truth table:

R2	p2	s2	C	D
0	0	0	0	0
0	0	1	1	1
0	1	0	1	0
0	1	1	0	0
1	0	0	0	1
1	0	1	0	0
1	1	0	0	0
1	1	1	0	0

3)C-map:

R2/P2 S2	00	01	11	10
0		1		1
1				

=R2`(P2 xorS2)

D-map:

R2/P2 S2	00	01	11	10
0		1		
1	1			

=P2(R2 xor S2)

Part 2

1) specification:

- Input = A,B (Player 1) C,D (Player 2)
- output= win 1 ,win 2,Tie

2)truth table:

A	B	C	D	W1	W2	T
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	0	0	0
0	0	1	1	0	0	0
0	1	0	0	0	0	0
0	1	0	1	0	0	1
0	1	1	0	0	1	0
0	1	1	1	1	0	0

1	0	0	0	0	0	0
1	0	0	1	1	0	0
1	0	1	0	0	0	1
1	0	1	1	0	1	0
1	1	0	0	0	0	0
1	1	0	1	0	1	0
1	1	1	0	1	0	0
1	1	1	1	0	0	1

3) W1-map:

AB/CD	00	01	11	10
00				
01			1	
11				1
10		1		

=BC(A xor D)+AB`C`D

W2-map:

AB/CD	00	01	11	10
00				
01				1
11		1		
10			1	

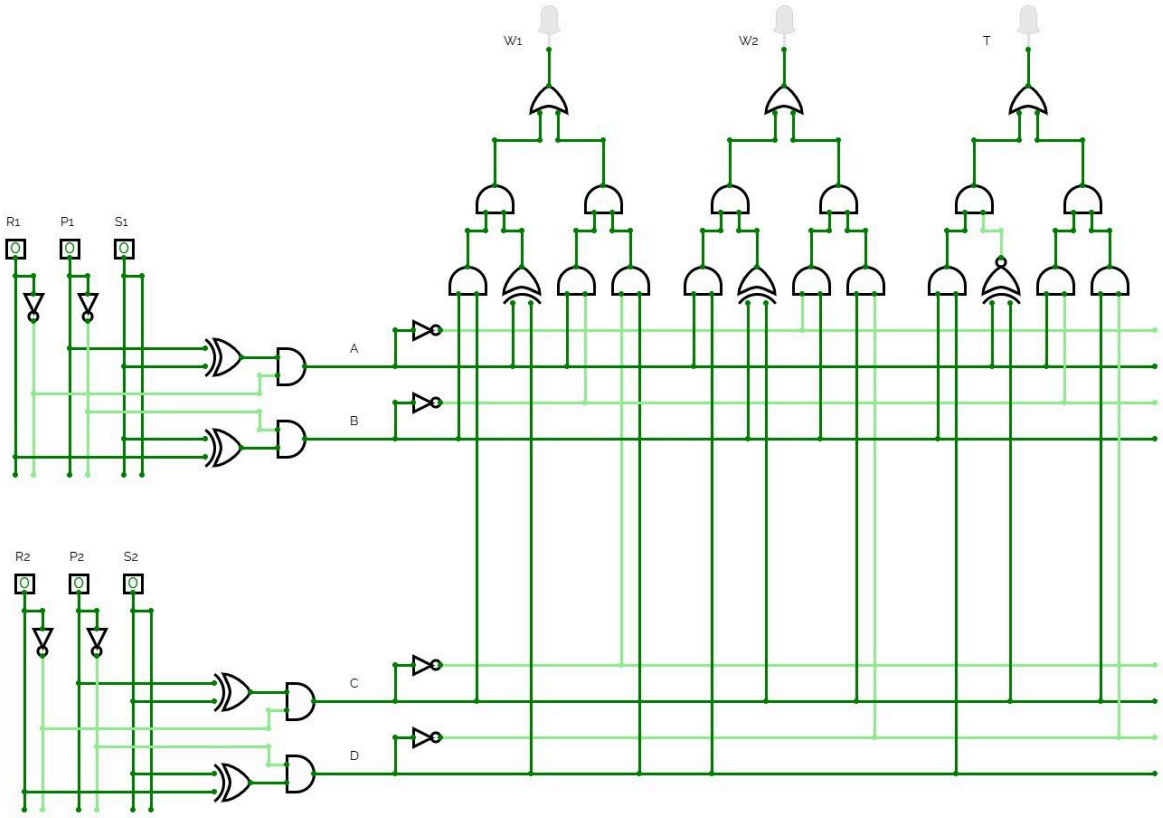
=AD(B xorC) + A`BCD`

T-map:

AB/CD	00	01	11	10
00				
01		1		
11			1	
10				1

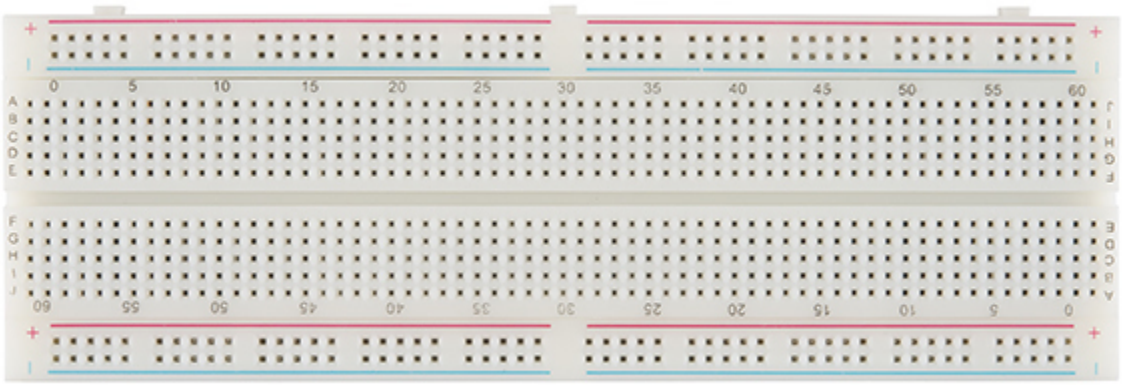
=BD(A xor c)` + AB`CD`

4)Desgin circuit:



MATERIALS

- Breadboard
  A breadboard is a tool for prototyping and testing electronic circuits without soldering . It allows components to be connected temporarily.



- Press (4pin)

A 4-pin press button (tactile switch) is primarily a momentary switch used as a standard input on electronic projects, such as for control panels, consumer electronics, and Arduino boards.



- IC :

AND GATE

An AND gate is used to perform logical Multiplication of binary input. The Output state of the AND gate will be high (1) if both the input is high (1), else the output state will be low(0) if any of the input is low (0).

OR GATE

OR GATE is most widely used digital logic circuit. The output state of OR gate will be high i.e., (1) if any of the input state is high or 1, else output state will be low i.e., 0.

NOT GATE

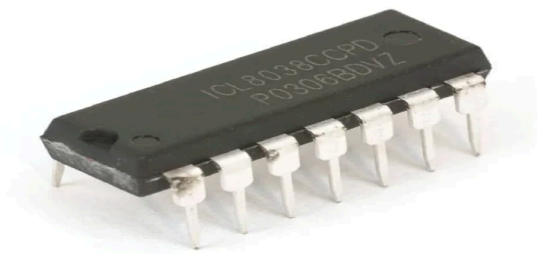
In digital electronics, the NOT gate is one of the basic Logic Gate having only a single input and a single output. It is also known as inverter or inverting buffer. When the input signal is "low" the output signal is "high" and vice-versa.

XOR GATE

It is used extensively in arithmetic logic circuits., logic comparators and error detection circuits. The XOR gate can take only two inputs at a time and give an output. The output of the XOR gate is high (1) only when its two inputs are dissimilar i.e., if one of them is low (0) then other one will be high (1).

XNOR GATE

The XNOR is the combination of XOR gate and NOT gate. The output of the XNOR gate is high(1) when both the inputs are high (1) or low(0). In other words, the output of the XNOR gate is high(1) when both the inputs are the same. the XNOR gate can sometimes be called as Equivalence gate.



Cost of circuit:

- BB-01 Breadboard 830 Tie Point x5 (175.00 EGP)
- IC 7404 - NOT Logic Inverter - Original x5 (75.00 EGP)
- IC 7408 - Quad 2 input AND Gates - Original x8 (120.00 EGP)
- IC 4077 - Quad Exclusive NOR XNOR Gate x1 (20.00 EGP)
- IC 4030 - Quad Exclusive OR XOR Gate x4 (72.00 EGP)
- IC 7432 - Quad 2 Input OR Gates - Original x3 (45.00 EGP)
- Press 4pin 6x6x4.5mm Tack Switch x6 (4.50 EGP) (Pushbuttons)
- Kit Resistors 1/4W -100 Carbon Resistors 1 % Tolerance 1/4W x1 (25.00 EGP)
- PH61 -20cm Male to Male 40 Jumper Wires Set x2 (70.00 EGP)
- LED 5mm Red Color x1 (0.50 EGP)
- LED 5mm Green Color x1 (0.50 EGP)
- LED 5mm Yellow Color x1 (0.50 EGP)

-Battery 1.5V AA Size - TOSHIBA x2 (20.00 EGP)

- Battery Case 2xAA On/Off Switch x1 (18.00 EGP)

Total: 646 EGP